

# Noise to Insight

AI Meets Data — Microsoft Build 2026

From messy exports to executive-ready intelligence in one pipeline

**Agentic Insights**

# Problem Statement

Data is everywhere — insight is rare

- Teams receive raw CSV/JSON exports (HR, orders, ops) with inconsistent schemas, missing dates, and hidden relationships.
- Manual cleaning and one-off dashboards take days; critical signals stay buried.
- Leaders need answers: *What patterns matter? Who is an outlier? What happens next?*
- Challenge: turn **noise** into **actionable insight** without a data-science team.

# Solution Overview

Six-phase agentic pipeline

- **Noise to Insight** — upload messy files; LangGraph orchestrates six phases end-to-end.
- Outputs per run: cleaned data, ranked insights, explained anomalies, forecast/snapshot chart, knowledge graph, executive report.
- **Streamlit Mission Control** — phase-by-phase tabs, KPIs, charts, downloads.
- Works with or without Azure OpenAI (statistical core + LLM enrichment).

# Architecture

Ingest → Clean → Discover → Detect → Predict → Graph → Report

| Phase       | Capability                         |
|-------------|------------------------------------|
| 0 Ingest    | Profile schema & quality           |
| 1 Clean     | LLM plan + Polars execution        |
| 2 Patterns  | Correlations & segment lift        |
| 3 Anomalies | Isolation Forest + LLM explain     |
| 4 Forecast  | Prophet / snapshot analytics       |
| 5 Graph     | Knowledge graph (NetworkX + PyVis) |
| 6 Report    | Executive HTML / PDF               |

Stack: Polars, DuckDB, scikit-learn, Prophet, NetworkX, LangGraph, Azure OpenAI, Streamlit

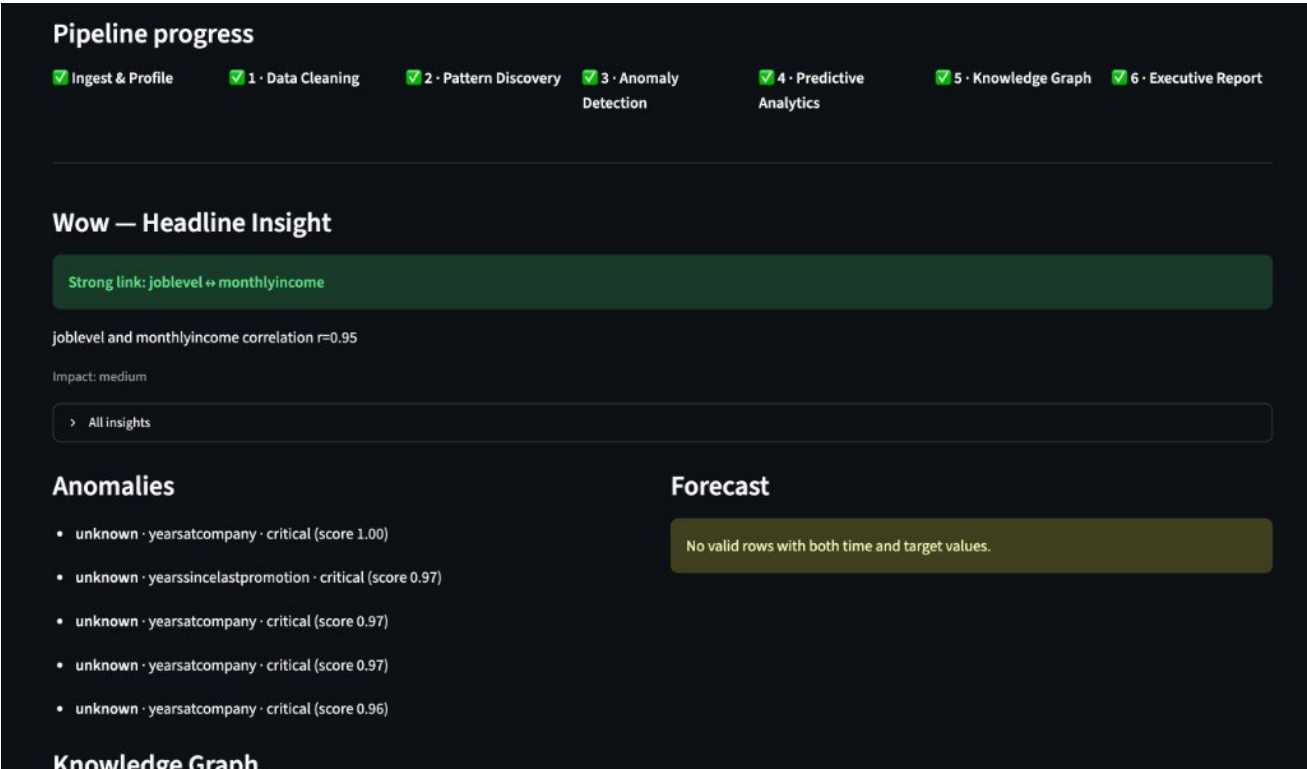
# AI Integration Details

Azure OpenAI at every decision point

| Phase     | AI role                                      | Fallback                             |
|-----------|----------------------------------------------|--------------------------------------|
| Cleaning  | LLM generates column rename/drop/impute plan | Heuristic schema normalization       |
| Patterns  | Ranked insight cards from statistics         | Template insights from correlations  |
| Anomalies | Hypothesis + recommended actions             | Isolation Forest + templates         |
| Forecast  | Narrative + prescriptive actions             | Prophet / sklearn + snapshot bars    |
| Graph     | Entity/relation extraction from samples      | Co-occurrence heuristic graph        |
| Report    | Executive summary synthesis                  | Jinja HTML + optional WeasyPrint PDF |

# Demo — Pipeline & Headline Insight

HR Employee Attrition dataset (1,470 rows)



# Demo — Anomaly Detection

Multivariate outliers with employee IDs & actions

**Critical — act now**

**unknown · yearsatcompany** (score 1.00)

Multivariate outlier: yearsatcompany deviates from peer behavior for unknown.

→ Investigate unknown — validate yearsatcompany and related operational events.

**unknown · yearssincepromotion** (score 0.97)

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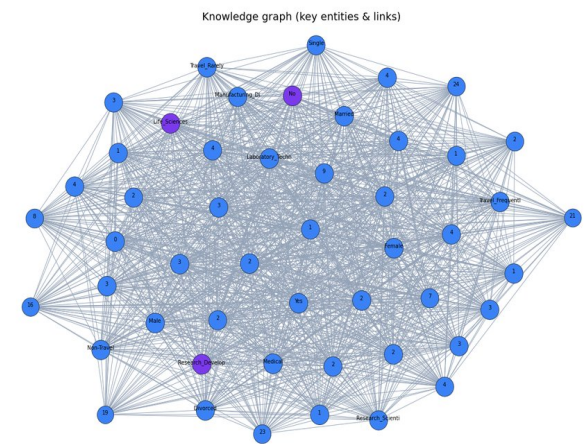
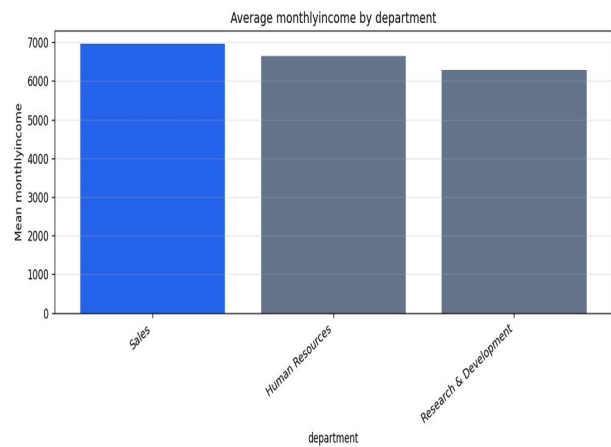
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# Demo — Forecast & Knowledge Graph

Snapshot analytics + entity network



Wow moment: JobLevel  $\leftrightarrow$  MonthlyIncome  $r \approx 0.95$ ; department income snapshot; connected HR entities

# Impact & Theme Alignment

“I had no idea that was in our data”

- **Intelligent cleaning** — normalize 35 HR columns automatically.
- **Pattern discovery** — surface strong correlations & segment lifts.
- **Actionable anomalies** — who to investigate and why (not just scores).
- **Analytics** — time-series forecast or snapshot bars when no dates exist.
- **Executive report** — one-click HTML/PDF for leadership.

# Team

|                         |                      |
|-------------------------|----------------------|
| <b>Team name</b>        | Agentic Insights     |
| <b>Team contributor</b> | Shiraptinath C R     |
| <b>Type</b>             | Solo Product Builder |
| <b>Project</b>          | Noise to Insight     |
| <b>Repository</b>       | Build                |

Thank you — Microsoft Build 2026 Hackathon